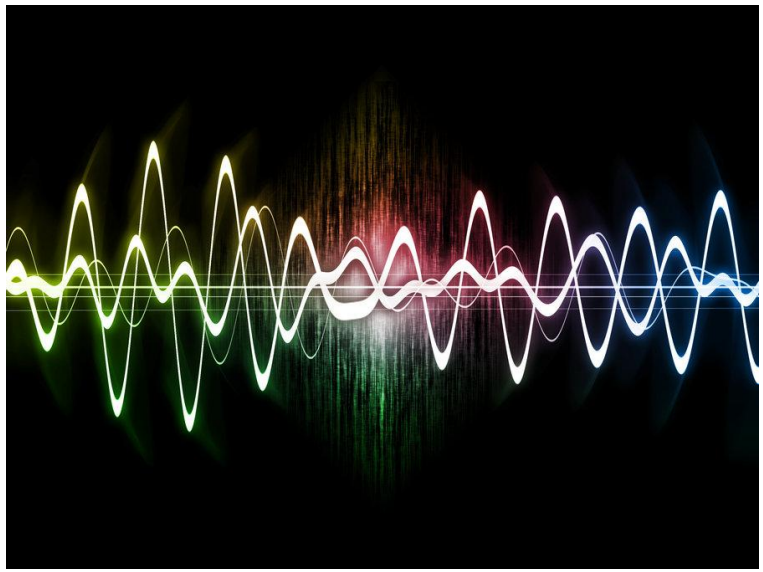
Week 2: Trigonometrics, powers, and vectors

- Trigonometric functions (sin, ...)
- Powers
- Vectors in 2 and 3 dimensions
- Lines and planes
- Distances, angles and the inner product
- The Cross Product

Sound



Sound, water, electro-magnetic and other waves are a sum of elementary waves: $A \sin(f \cdot 2\pi \cdot t)$

- $\sum_{k=1}^{n} A_k \sin(2\pi \cdot f_k \cdot t)$

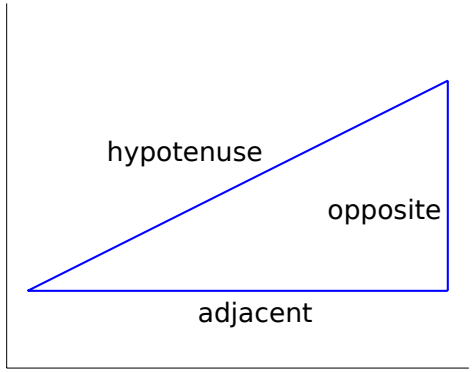
where the frequency f determines the pitch and
the amplitude A the intensity of the sound. sin is the sine function

Trigonometric functions: sin, cos, tan

$\sin = \frac{\text{opp}}{\text{hyp}}$, $\cos = \frac{\text{adj}}{\text{hyp}}$, $\tan = \frac{\text{opp}}{\text{adj}}$

Relation degrees – radians:

$30^\circ = \frac{\pi}{6}$, $45^\circ = \frac{\pi}{4}$, $60^\circ = \frac{\pi}{3}$, $90^\circ = \frac{\pi}{2}$



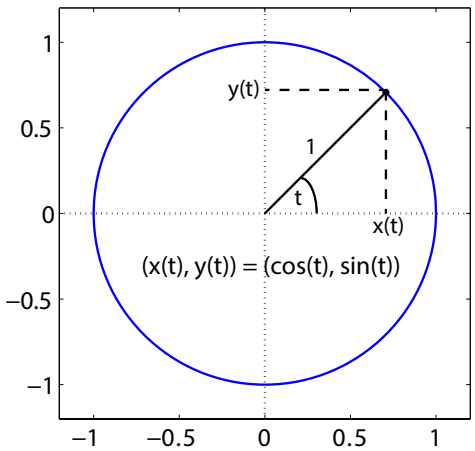
Radians = the length of a piece of the circle

Complete circle $= 2\pi = 360^\circ$

Point on a circle with radius 1 has the coordinates $(\cos(t), \sin(t))$

This agrees with $\sin^2(t) + \cos^2(t) = (\sin(t))^2 + (\cos(t))^2 = 1$

Similar: point on the circle with radius a has the coordinates $(a\cos(t), a\sin(t)) = a(\cos(t), \sin(t))$

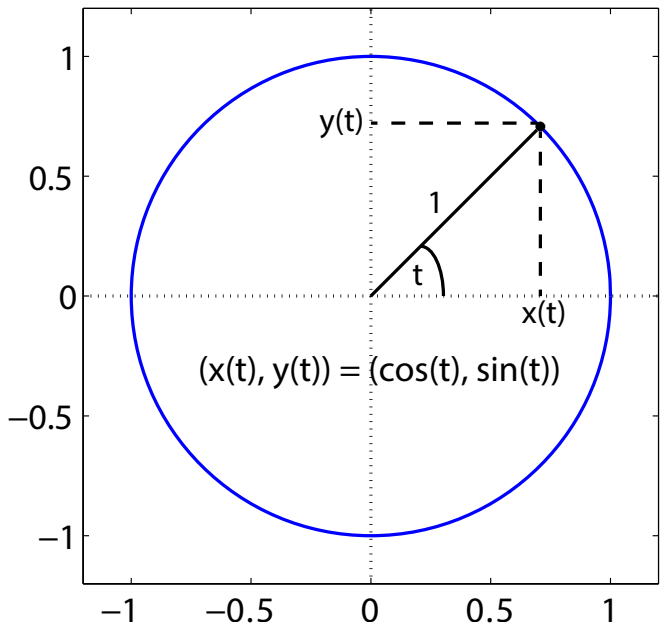


The 4 quadrants

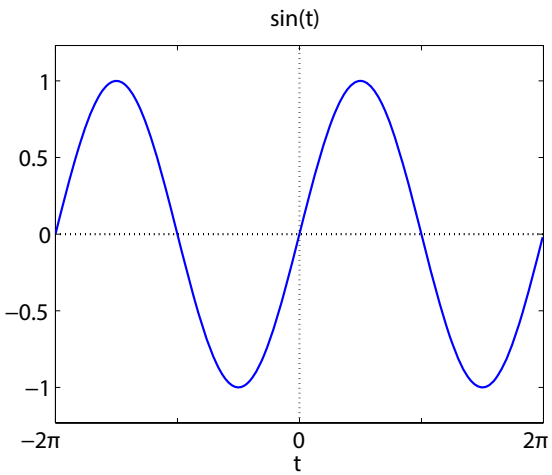
1. $t \in [0, \pi/2] \Longrightarrow (cos(t), sin(t))$ is in the 1st quadrant
2. $t \in [\pi/2, \pi] \Longrightarrow (cos(t), sin(t))$ is in the 2nd quadrant
3. $t \in [\pi, 3/2 \cdot \pi] \Longrightarrow (cos(t), sin(t))$ is in the 3rd quadrant
4. $t \in [3/2 \cdot \pi, 2\pi] \Longrightarrow (cos(t), sin(t))$ is in the 4th quadrant

so that we define

1. the 1st quadrant: $t \in [0, \pi/2]$
2. the 2nd quadrant: $t \in [\pi/2, \pi]$
3. the 3rd quadrant: $t \in [\pi, 3/2 \cdot \pi]$
4. the 4th quadrant: $t \in [3/2 \cdot \pi, 2\pi]$



Sine, Cosine, and Tangens functions



$\sin(t + 2\pi) = \sin(t)$

odd function:

$\sin(-t) = -\sin(t)$

$\sin(t) = 0$

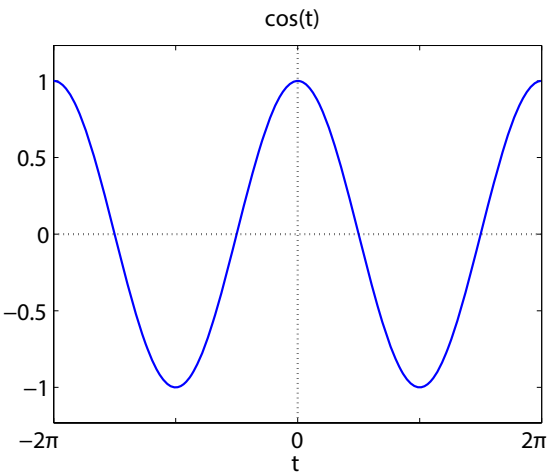
$\Longrightarrow \quad t = k\pi \qquad (k \in \mathbb{Z})$

$\sin(t) = 1$

$\Longrightarrow \quad t = \frac{\pi}{2} + 2k\pi$

$\sin(t) = -1$

$\Longrightarrow \quad t = \frac{3\pi}{2} + 2k\pi$



$\cos(t + 2\pi) = \cos(t)$

even function:

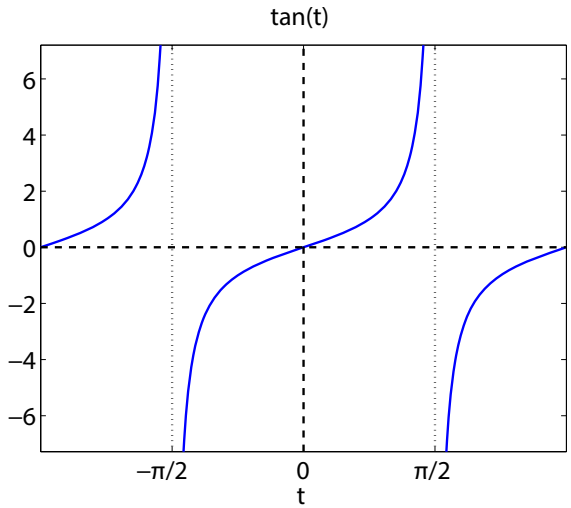
$\cos(-t) = \cos(t)$

$\cos(t) = 0$

$\Longrightarrow \quad t = \frac{\pi}{2} + k\pi$

$\sin(t) = \cos(\frac{\pi}{2} - t)$

$\cos(t) = \sin(\frac{\pi}{2} - t)$



$\tan(t) = \frac{\sin(t)}{\cos t}$

odd function:

$\tan(-t) = -\tan(t)$

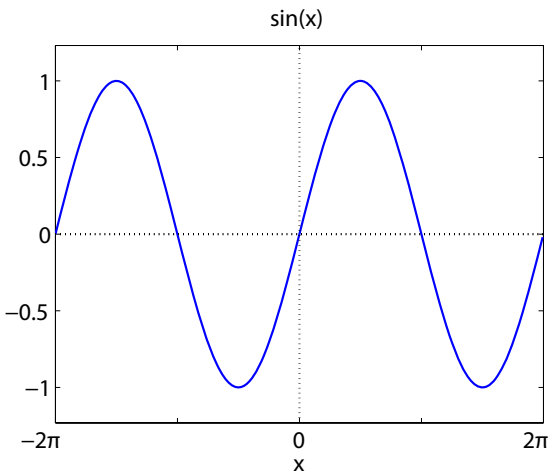
$\tan(t) = 0$

$\Longrightarrow \quad t = k\pi \qquad (k \in \mathbb{Z})$

vertical asymptote in

$\frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$

Sine, Cosine, and Tangens functions



$\sin(x + 2\pi) = \sin(x)$

odd function:

$\sin(-x) = -\sin(x)$

$\sin(x) = 0$

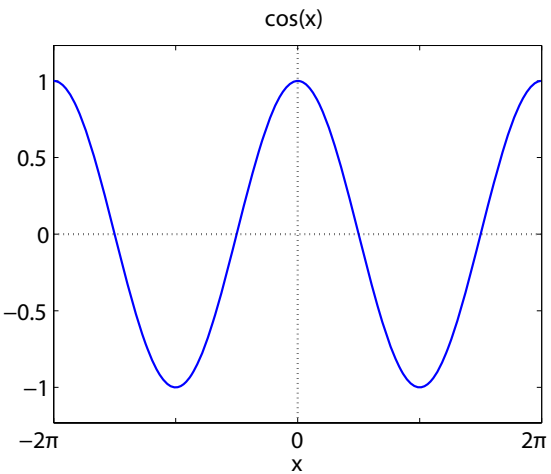
$\Longrightarrow \quad x = k\pi \qquad (k \in \mathbb{Z})$

$\sin(x) = 1$

$\Longrightarrow \quad x = \frac{\pi}{2} + 2k\pi$

$\sin(x) = -1$

$\Longrightarrow \quad x = \frac{3\pi}{2} + 2k\pi$



$\cos(x + 2\pi) = \cos(x)$

even function:

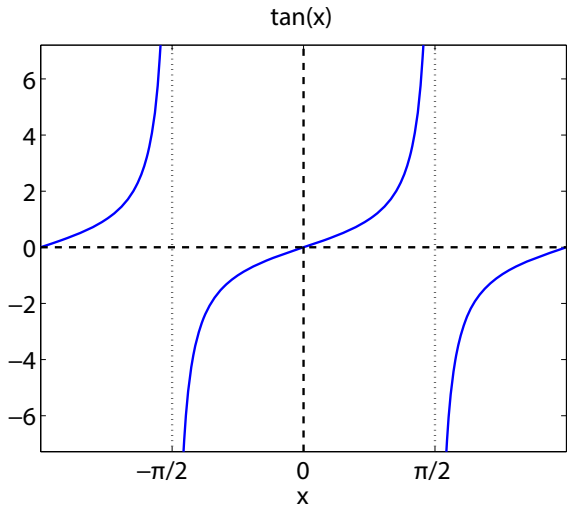
$\cos(-x) = \cos(x)$

$\cos(x) = 0$

$\Longrightarrow \quad x = \frac{\pi}{2} + k\pi$

$\sin(x) = \cos(\frac{\pi}{2} - x)$

$\cos(x) = \sin(\frac{\pi}{2} - x)$



$\tan(x) = \frac{\sin(x)}{\cos x}$

odd function:

$\tan(-x) = -\tan(x)$

$\tan(x) = 0$

$\Longrightarrow \quad x = k\pi \qquad (k \in \mathbb{Z})$

vertical asymptote in

$\frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}$

Trigonometric identities

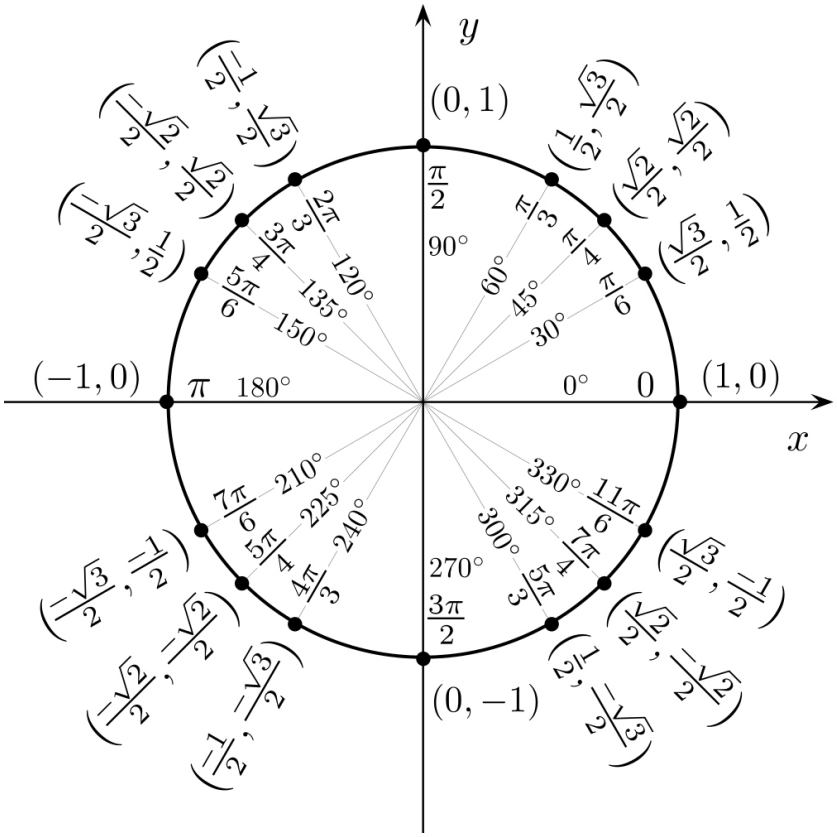
Know at least the red ones by heart!

$$
\begin{aligned}
\sin^2(x) + \cos^2(x) &= 1 \\
\sin(x+y) &= \sin(x)\cos(y) + \cos(x)\sin(y) \\
\sin(2x) &= 2\sin(x)\cos(x) \\
\cos(x+y) &= \cos(x)\cos(y) - \sin(x)\sin(y) \\
\cos(2x) &= \cos^2(x) - \sin^2(x) \\
&= 2\cos^2(x) - 1 \\
&= 1 - 2\sin^2(x)
\end{aligned}
$$

Furthermore, we can use the last two lines to rewrite $\sin^2(x)$ and $\cos^2(x)$

$$\sin^2(x) = \tfrac{1}{2} - \tfrac{1}{2}\cos(2x) \qquad \cos^2(x) = \tfrac{1}{2} + \tfrac{1}{2}\cos(2x)$$

Angles on the unit circle



Do we need to remember all of those?!?

No, just those in the 1st quandrant:

x	0	$\frac{1}{6}\pi$	$\frac{1}{4}\pi$	$\frac{1}{3}\pi$	$\frac{1}{2}\pi$
$\sin(x)$	$\frac{1}{2}\sqrt{0}$	$\frac{1}{2}\sqrt{1}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{4}$
$\cos(x)$	$\frac{1}{2}\sqrt{4}$	$\frac{1}{2}\sqrt{3}$	$\frac{1}{2}\sqrt{2}$	$\frac{1}{2}\sqrt{1}$	$\frac{1}{2}\sqrt{0}$
$\tan(x)$	0	$\frac{1}{3}\sqrt{3}$	1	$\sqrt{3}$	*ng*

$$
\begin{aligned}
\sin(\pi - x) &= \sin(\pi + (-x)) \\
&\overset{\sin(x+y)=\ldots}{=} \sin(\pi)\cos(-x) + \cos(\pi)\sin(-x) \\
&= 0 \cdot \cos(-x) + -1 \cdot \sin(-x) \\
&= \sin(x)
\end{aligned}
$$

Determining $\sin(x)$, $\cos(x)$, $\tan(x)$, $x \in [\frac{\pi}{2}, 2\pi]$ (I)

Use the formulas (that you know by heart)

- $\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y)$
- $\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y)$

one can relate angles in Q2-4 to one in Q1!

Very practical: use π and 2π!

$\sin(x) = \sin(\pi - x)$	$\cos(x) = -\cos(\pi - x)$	$\tan(x) = -\tan(\pi - x)$
$\sin(x) = -\sin(x - \pi)$	$\cos(x) = -\cos(x - \pi)$	$\tan(x) = \tan(x - \pi)$
$\sin(x) = -\sin(-x)$	$\cos(x) = \cos(-x)$	$\tan(x) = -\tan(-x)$

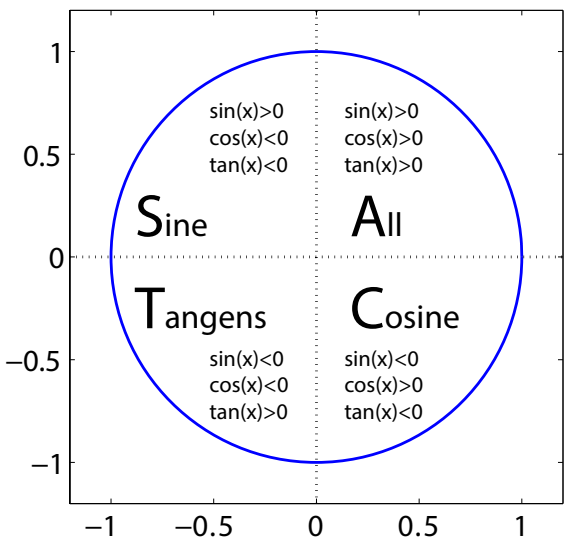
Ex: Calculate $\sin(x)$ for $x = \frac{5}{6}\pi$.
$\frac{5}{6}\pi$ is in the 2nd quadrant and
$\pi - \frac{5}{6}\pi$ in the 1st: $\sin(\frac{5}{6}\pi) = \sin(\pi - \frac{5}{6}\pi) = \sin(\frac{1}{6}\pi) = \frac{1}{2}$.

Determining $\sin(x)$, $\cos(x)$, $\tan(x)$, $x \in [\frac{\pi}{2}, 2\pi]$ (II)

Who can (and wants to) remember all 9 relations?
Instead, remember to change the argument in the trigonometric function according to:

- if x is in 2nd quadrant $[\frac{1}{2}\pi, \pi]$: $x \to \pi - x$
- if x is in 3rd quadrant $[\pi, \frac{3}{2}\pi]$: $x \to x - \pi$
- if x is in 4th quadrant $[\frac{3}{2}\pi, 2\pi]$: $x \to 2\pi - x$

and the CAST rule to check for a changing sign.



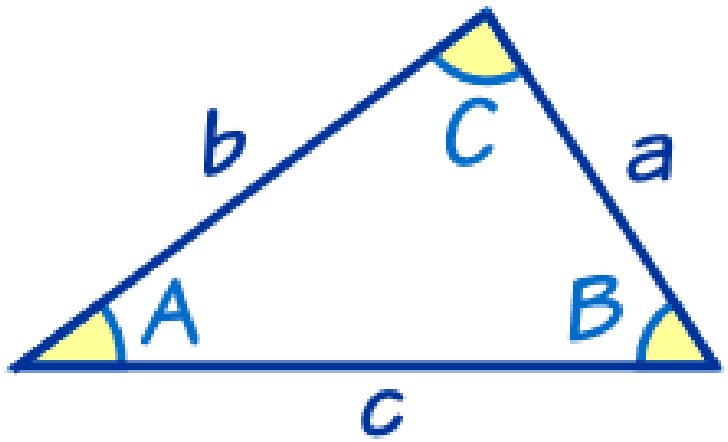
Ex: Calculate $\sin(x)$ for $x = \frac{7}{6}\pi$.
$\frac{7}{6}\pi$ is in the 3rd quadrant.
$\frac{7}{6}\pi \to \frac{7}{6}\pi - \pi = \frac{1}{6}\pi$
$sin(x)$ changes sign from Q3 to Q1
so that $\sin(\frac{7}{6}\pi) = -\sin(\frac{7}{6}\pi - \pi) = -\sin(\frac{1}{6}\pi) = -\frac{1}{2}$.

Determining $\sin(x)$, $\cos(x)$, $\tan(x)$, $x \in [0, 2\pi]$ (III)

$$
\begin{aligned}
\sin(\tfrac{\pi}{6}) &= \tfrac{1}{2} \\
\cos(\tfrac{2}{3}\pi) &= -\cos(\tfrac{1}{3}\pi) &&= -\tfrac{1}{2} && \text{qu 2, } \cos(\pi - x) \\
\tan(-\tfrac{\pi}{4}) &= -\tan(\tfrac{\pi}{4}) &&= -1 && \text{qu 1, tan is odd} \\
\sin(\tfrac{4\pi}{3}) &= -\sin(\tfrac{\pi}{3}) &&= -\tfrac{1}{2}\sqrt{3} && \text{qu 3, } \sin(x - \pi) \\
\cos(\tfrac{7}{6}\pi) &= -\cos(\tfrac{1}{6}\pi) &&= -\tfrac{1}{2}\sqrt{3} && \text{qu 3, } \cos(x - \pi) \\
\tan(\tfrac{11}{6}\pi) &= -\tan(\tfrac{1}{6}\pi) &&= -\tfrac{1}{3}\sqrt{3} && \text{qu 4, } \tan(x - 2\pi)
\end{aligned}
$$

Do [A P.7 T5]

Sine and Cosine Rules



Sine rule

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Cosine rule

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc\cos A \\ b^2 &= c^2 + a^2 - 2ca\cos B \\ c^2 &= a^2 + b^2 - 2ab\cos C \end{aligned}$$

Do [A P.7 T31,37].

Trigonometric Equations and Inequalities

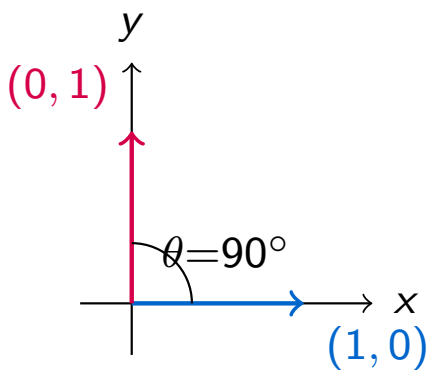
$$
\begin{aligned}
\sin(x) = \sin(a) &\implies x = a + 2k\pi \text{ or } x = \pi - a + 2k\pi \\
\cos(x) = \cos(a) &\implies x = \pm a + 2k\pi \\
\tan(x) = \tan(a) &\implies x = a + k\pi \\
\sin(x) = \cos(a) &\implies \text{rewrite cos to sin :} \\
&\qquad \cos(a) \underset{\sin(x+y)=\ldots}{=} \sin\left(\tfrac{\pi}{2} - a\right)
\end{aligned}
$$

Recipe: Trigonometric Equations/Inequalities

1. Use substitution to rewrite to an equation with (powers of) only 1 unknown, eg $\sin(x)$
   and with only 1 argument (so no mix of $\sin(x)$ and $\sin(2x)$)
   or products of different functions (avoid roots as much as possible).
2. Maybe use substitution, eg $y = \sin(x)$, and solve the equation / inequality
3. Rewrite y back to eg $\sin(x)$, and solve, eg
   $\sin(x) = \frac{1}{2} \Longrightarrow x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$
4. Pay attention to "$+k\,2\pi$" (sin and cos) or "$+k\,\pi$" for tan

Rotating a sprite on screen

A 2D game needs to rotate a character (a set of points) by an angle θ around the origin.
A point (x, y) rotated by angle θ moves to (x', y') with



$$\boxed{x' = x\cos(\theta) - y\sin(\theta),\ \ y' = x\sin(\theta) + y\cos(\theta)}$$

Example: rotate $(1, 0)$ by $\theta = 90° = \frac{\pi}{2}$:

$$x' = 1 \cdot \cos\left(\tfrac{\pi}{2}\right) - 0 \cdot \sin\left(\tfrac{\pi}{2}\right) = 0, \qquad y' = 1 \cdot \sin\left(\tfrac{\pi}{2}\right) + 0 \cdot \cos\left(\tfrac{\pi}{2}\right) = 1$$

so $(1, 0) \to (0, 1)$ – exactly a quarter turn, as shown.
Every rotated sprite, spinning icon, or clock hand you have ever seen on a screen is built from sin and cos exactly like this.

Spinning Sonic



- Hardware Constraint: Sega Genesis CPU simply wasn't fast enough to calculate matrix math or trigonometry to rotate an image in real time.
- "Sprite Sheets" Trick: Instead of coding rotation, artists created a grid of drawings (a sprite sheet) for every single angle needed—like Sonic running at 45 degrees up a hill or tucked into a tight 90-degree ball.
- Memory vs. Math: Early game development was a constant trade-off. It was vastly cheaper for the hardware to pull a pre-calculated picture from the game cartridge's memory than to calculate a mathematical rotation on the fly.

Exercises

1. For $\sin(x) = -\frac{5}{13}$ and $x \in [3/2\pi, 2\pi]$, determine $\cos(x)$ and $\tan(x)$.

2. For $\tan(x) = -\frac{7}{24}$, and x in the 4th quadrant, determine $\sin(x)$ and $\cos(x)$.

3. Calculate $\sin(\frac{5\pi}{12})$. Hint: make use of the $\sin(x + y)$ identity.

4. Solve $3\cos^2(x) - \sin^2(x) = 0$ for $x \in (0, 2\pi)$.

5. Solve $\cos^2(x) + \cos(x) = \sin^2(x)$.

Exercises

1. For $\sin(x) = -\frac{5}{13}$ and $x \in [3/2\pi, 2\pi]$, determine $\cos(x)$ and $\tan(x)$.
   $\cos(x) = \frac{12}{13}$ and $\tan(x) = -\frac{5}{12}$
2. For $\tan(x) = -\frac{7}{24}$, and x in the 4th quadrant, determine $\sin(x)$ and $\cos(x)$.
   $\cos(x) = \frac{24}{25}$ and $\sin(x) = -\frac{7}{25}$
3. Calculate $\sin(\frac{5\pi}{12})$. Hint: make use of the $\sin(x + y)$ identity.
   $\frac{1}{4}(\sqrt{6} + \sqrt{2})$
4. Solve $3\cos^2(x) - \sin^2(x) = 0$ for $x \in (0, 2\pi)$.
   $x = \frac{\pi}{3} \vee x = \frac{4\pi}{3} \vee x = \frac{2\pi}{3} \vee x = \frac{5\pi}{3}$
5. Solve $\cos^2(x) + \cos(x) = \sin^2(x)$.
   $x = \pi + 2n\pi \vee x = \frac{\pi}{3} + 2n\pi \vee x = \frac{5\pi}{3} + 2n\pi$ with $n \in \mathbb{Z}$

Powers and roots

Reminder: Rules for calculating powers and roots

$$a^p a^q = a^{p+q} \qquad (ab)^p = a^p b^p \qquad a^{-p} = \frac{1}{a^p}$$

$$a^{1/2} = \sqrt{a} \qquad a^{1/n} = \sqrt[n]{a} \qquad a^{\frac{p}{q}} = \sqrt[q]{a^p}$$

$$(a^p)^q = a^{pq} \qquad a^{-\frac{p}{q}} = \frac{1}{a^{\frac{p}{q}}} = \frac{1}{\sqrt[q]{a^p}} \qquad \left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$$

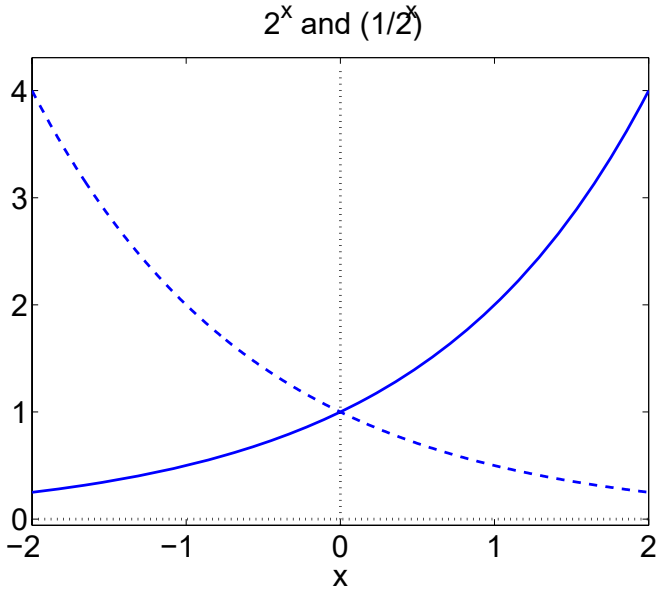
Useful to rewrite expressions into more simplified forms:

Ex: $\dfrac{(3a^{-\frac{2}{3}}b)^2}{2a^2b^{-\frac{1}{3}}} = \dfrac{3^2(a^{-\frac{2}{3}})^2b^2}{2a^2b^{-\frac{1}{3}}} = \dfrac{9a^{-\frac{4}{3}}b^2}{2a^2b^{-\frac{1}{3}}} = \dfrac{9}{2}a^{-\frac{10}{3}}b^{\frac{7}{3}}$

Equations with powers

Rewrite to

$$a^{\text{expression}_1} = a^{\text{expression}_2}$$



conclude that the exponents must be equal because:

- $f(x) = a^x$ everywhere increasing function if $a > 1$
- $f(x) = a^x$ everywhere decreasing function if $0 < a < 1$

Ex:

$$2^{x+1} + 2^{x+3} = 320 \implies 2^x(2 + 2^3) = 320 \implies 2^x = 32 = 2^5 \implies x = 5$$

Ex: $16^{3x+3} = 8^{x^2+4} \implies (2^4)^{3x+3} = (2^3)^{x^2+4} \implies 2^{12x+12} = 2^{3x^2+12}$

$$\implies 12x + 12 = 3x^2 + 12 \implies 3x(x-4) = 0 \implies x = 0 \text{ or } x = 4$$

Inequalities with powers

Again: get the same base number, then compare the powers
Remember:

- $f(x) = a^x$ is increasing everywhere if $a > 1$
- $f(x) = a^x$ is decreasing everywhere if $0 < a < 1$

Ex: the base number: is it < 1 or > 1?

- $f(x) = 2^x$ is increasing everywhere because $2 > 1$
- $f(x) = 2^{-x} = (2^{-1})^x = (\frac{1}{2})^x$ is decreasing everywhere because $0 < 1/2 < 1$

Ex: Inequalities

- $2^x > 2^4 \implies x > 4$ because 2^x is increasing
- $(\frac{1}{2})^x > (\frac{1}{2})^4 \implies x < 4$ because $(\frac{1}{2})^x$ is decreasing

Quiz: Powers

1. Solve $e^{x^2} = 5e^{2x-1}$.

2. Solve $\frac{2^{2x}-8}{2^x-4} \leq 0$

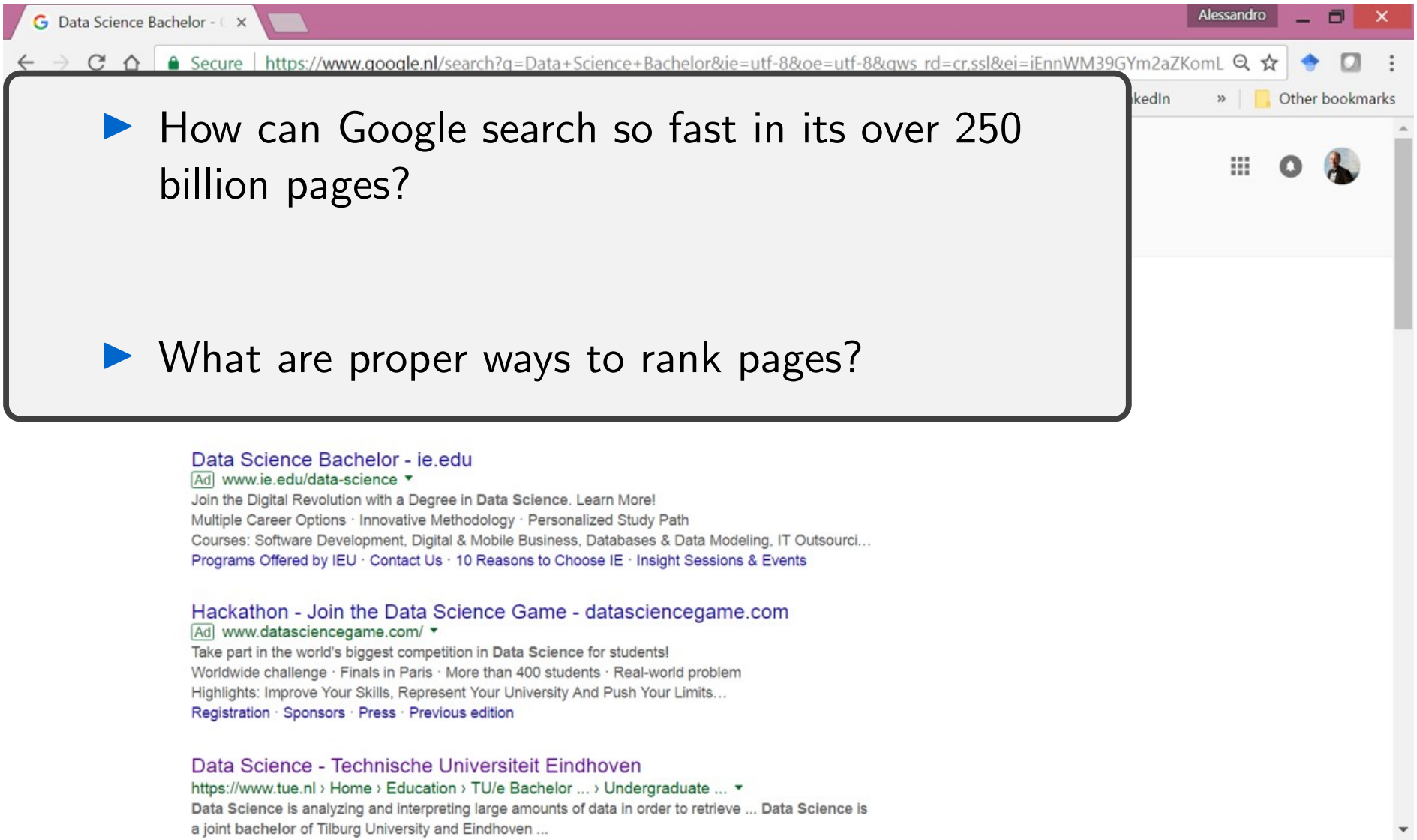
3. $f(x) = 5^x$ is
   (a) increasing everywhere
   (b) decreasing everywhere
   (c) increasing for $x > 0$, decreasing for $x < 0$
   (d) increasing for $x < 0$, decreasing for $x > 0$

4. If $x < 5$, then
   (a) $5^5 > 5^x$ (b) $5^x \leq 5^5$ (c) $5^5 < 5^x$ (d) $5^x \geq 5^5$

Quiz: Powers

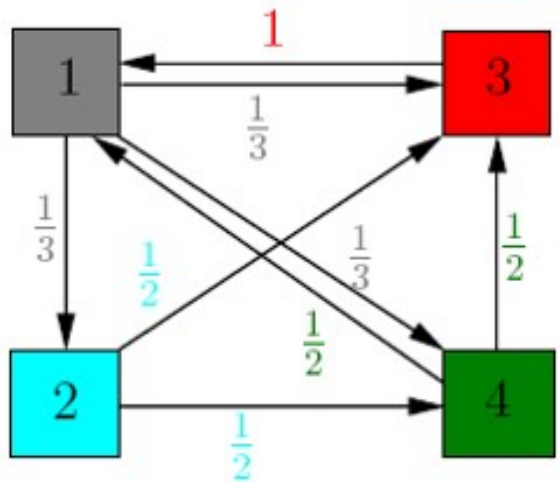
1. Solve $e^{x^2} = 5e^{2x-1}$.
   $x = 1 + \pm\sqrt{\ln(5)}$
2. Solve $\frac{2^{2x}-8}{2^x-4} \leq 0$
   $\frac{3}{2} \leq x < 2$
3. $f(x) = 5^x$ is
   (a) increasing everywhere
   (b) decreasing everywhere
   (c) increasing for $x > 0$, decreasing for $x < 0$
   (d) increasing for $x < 0$, decreasing for $x > 0$
4. If $x < 5$, then
   (a) $5^5 > 5^x$ (b) $5^x \leq 5^5$ (c) $5^5 < 5^x$ (d) $5^x \geq 5^5$

Google Search



- How can Google search so fast in its over 250 billion pages?
- What are proper ways to rank pages?

Mathematical Model for Weblinks



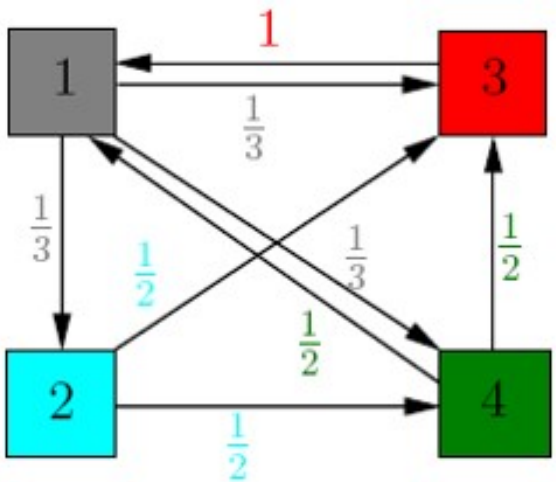
Model the WWW as a **graph**:

- Squares (nodes, vertices) denote web pages
- Arrows (edges) denote links from one page to another

Relevance is translated into number of incoming links weighted by the importance of referring pages.

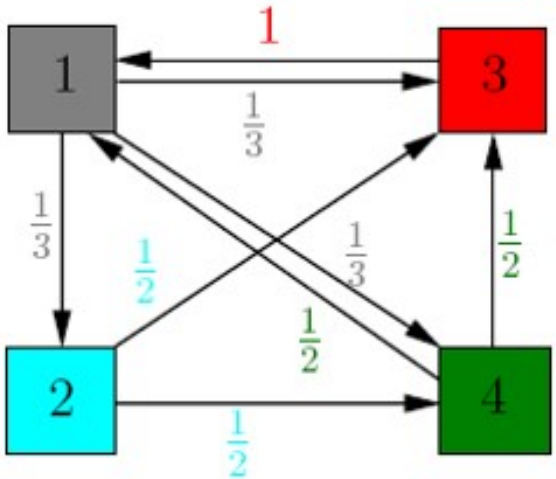
(What comes first?)

System of Linear Equations



$$
\begin{aligned}
x_1 &= 1 \cdot x_3 + \frac{1}{2} \cdot x_4 \\
x_2 &= \frac{1}{3} \cdot x_1 \\
x_3 &= \frac{1}{3} \cdot x_1 + \frac{1}{2} \cdot x_2 + \frac{1}{2} \cdot x_4 \\
x_4 &= \frac{1}{3} \cdot x_1 + \frac{1}{2} \cdot x_2
\end{aligned}
$$

System of Linear Equations - Matrix and Vectors



$$\begin{bmatrix} 0 & 0 & 1 & \frac{1}{2} \\ \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{2} & 0 & 0 \end{bmatrix} = A$$

$$A \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \frac{1}{31} \cdot \begin{bmatrix} 12 \\ 4 \\ 9 \\ 6 \end{bmatrix}$$

Information about page ranks is determined as a **vector** of page relevance (solutions of linear systems).

Back to the opening question: the ranking $\frac{1}{31}(12, 4, 9, 6)$ says page 1 is most relevant.

More Matrix-Vector problems





- User preferences/histories are encoded as vector data.
- How similar is your vector to those of others?
- completely similar = "parallel"
- completely different = "perpendicular"
- Trigonometric functions (just done) used with vector calculus (coming up)

Introduction: Vectors in Analytic Geometry

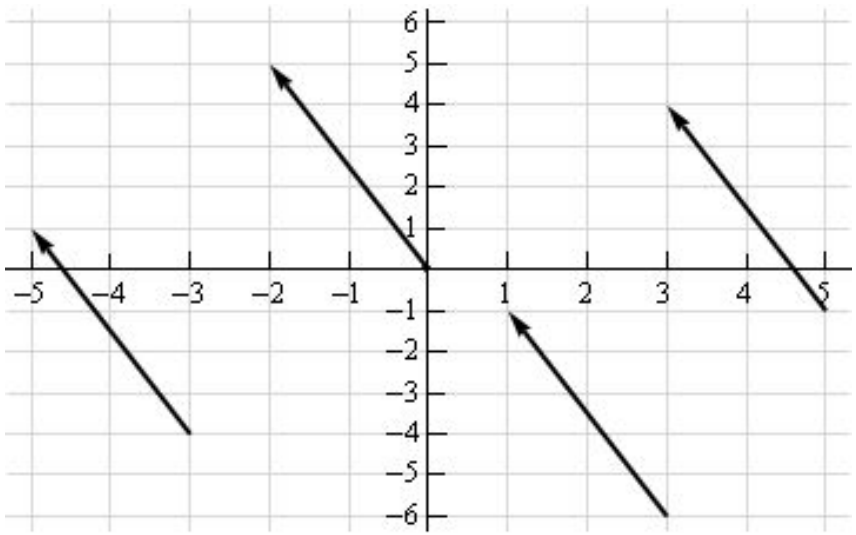
We will introduce vectors and their calculus in a specific application: Analytic Geometry

- describe directions in two or three space dimensions
- addition and scaling of vectors
- linear combinations
- lines and line segments in $\mathbb{R}^2$ and $\mathbb{R}^3$
- Planes in $\mathbb{R}^3$
- Orthogonality, angle between vectors
- Intersections of lines, lines and planes, planes

Vectors in 2 and 3 dimensions

A vector is represented in plane or space by an arrow with direction and length (magnitude).

Vector $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ (2 to the left, 5 up)



These vectors are equivalent, even though they have different starting points.

Vectors in 2 and 3 dimensions: Notation

In a plane (2D): 2 to the left and 5 up is a vector:

- Notation: Write $\begin{pmatrix} -2 \\ 5 \end{pmatrix}$ or $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ (lectures: latter notation)
- Sometimes "also": $<-2, 5>$: eg in books to save space...
- $\mathbf{v}$, $\underline{v}$, $\vec{v}$, or simply v (on the slides $\mathbf{v}$, on paper, I write v of $\underline{v}$)
- Sketch: Arrow (a combination of a line segment and directions)

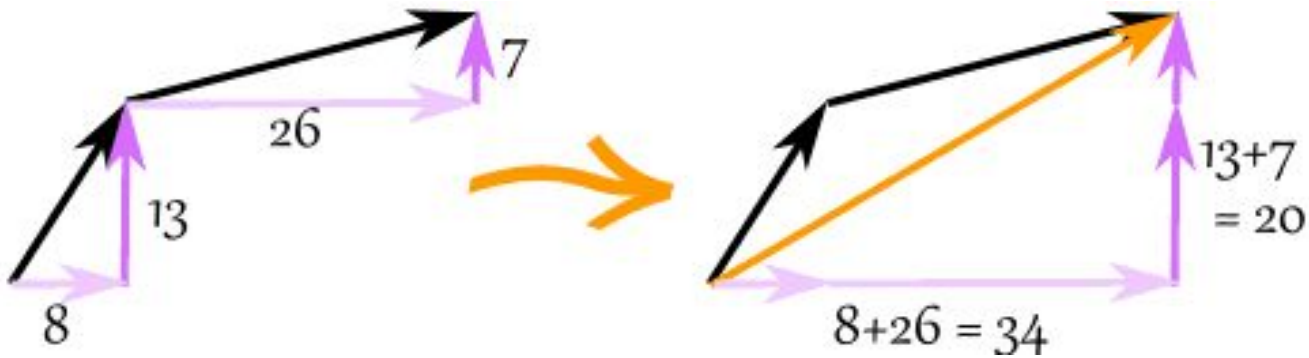
A vector has a direction and length (here $\sqrt{(-2)^2 + 5^2}$).

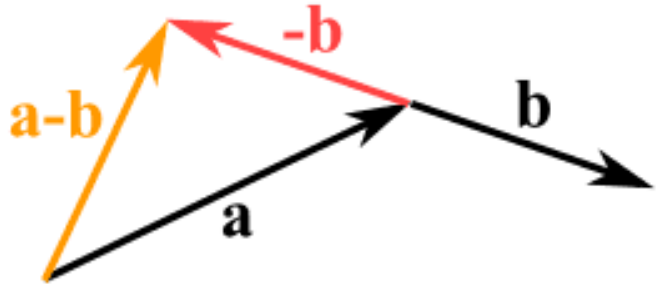
Vectors: Addition and subtraction

- add or subtract component-wise:

$$\begin{bmatrix} 8 \\ 13 \end{bmatrix} + \begin{bmatrix} 26 \\ 7 \end{bmatrix} = \begin{bmatrix} 34 \\ 20 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$$

- graphically: parallellogram





Vectors from point A to point B

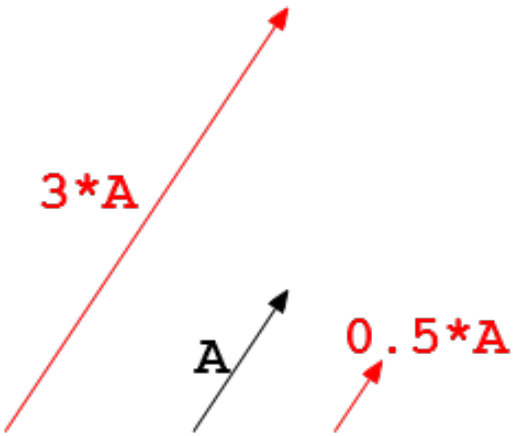
Because $A = (x_1, y_1)$ and $B = (x_2, y_2)$ then the vector from A to B is:

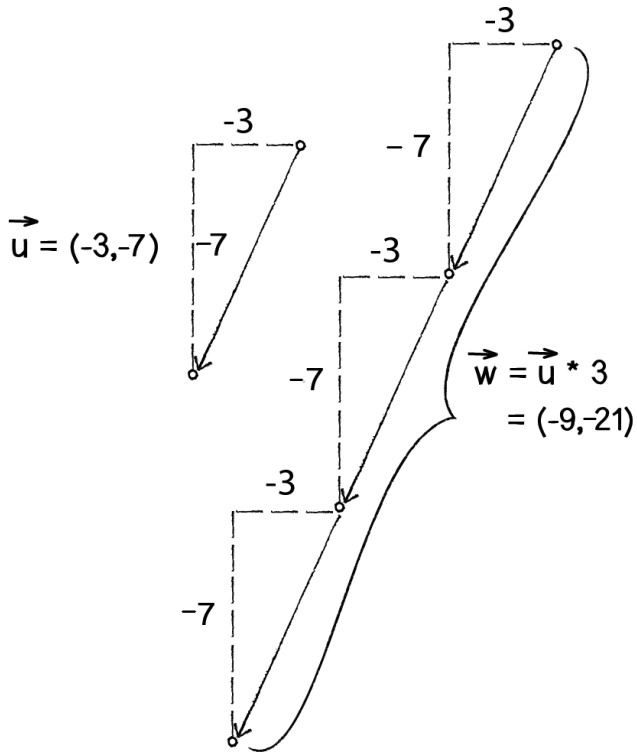
$$B - A = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} - \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

Notation: AB or $\vec{AB}$.
Do [V 5 T1]

Scaling of vectors

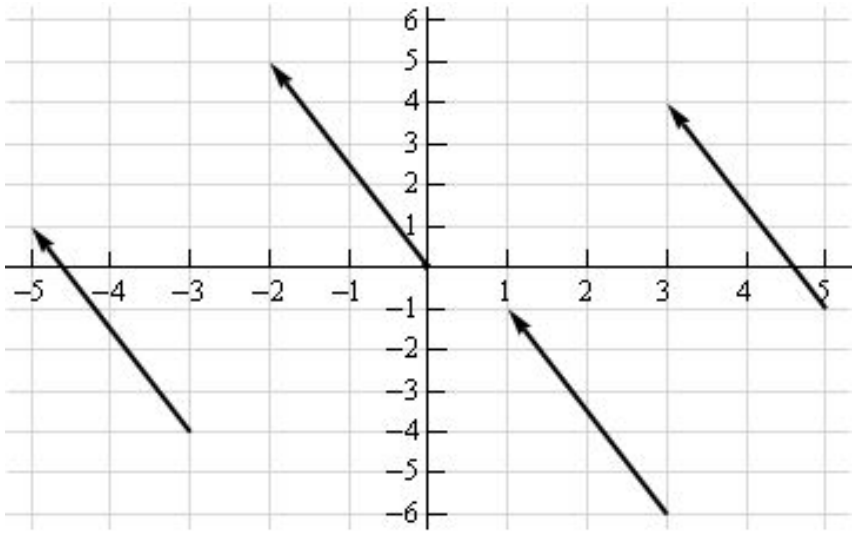
Scaling (scalar multiplication) changes the length and evt the direction:





- scalar means using (real) number
- multiply component-wise: $2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$

Scaling of vectors



- multiplication by 2:
  vector becomes 2 times as long, keeps the same direction
- multiplication by -2:
  vector becomes 2 times as long, in opposite direction

(Later for more about length)

Subtraction of vectors

Convention: Subtraction is addition of the vector with opposite direction:

$$\mathbf{b} - \mathbf{a} = \mathbf{b} + ((-1) \cdot \mathbf{a})$$

Ex: $\begin{bmatrix} 1 \\ -2 \end{bmatrix} - \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + (-1) \cdot \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix}$

Length of a vector

Notation:

- $\|\mathbf{x}\|$ (chosen)
- or sometimes also $|\mathbf{x}|$

Calculate: if $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ then

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} \quad \text{thus} \quad \|\mathbf{x}\|^2 = x_1^2 + x_2^2 + \cdots + x_n^2$$

Distance between two points $\mathbf{x}$ and $\mathbf{y}$: $\|\mathbf{x} - \mathbf{y}\|$

Ex: Distance between $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \sqrt{5}$

is also the length of the difference vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\left(\text{and } \begin{bmatrix} -1 \\ -2 \end{bmatrix}\right)$

Length of a vector

Ex: $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ has length $\sqrt{2}$

Sometimes we seek a vector in the same direction, but with length 1

Ex: $\begin{bmatrix} 1 \\ 1 \end{bmatrix} / \sqrt{2} = \begin{bmatrix} \frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} \end{bmatrix}$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \sqrt{2} \cdot \underbrace{\begin{bmatrix} \frac{1}{2}\sqrt{2} \\ \frac{1}{2}\sqrt{2} \end{bmatrix}}_{\text{length 1}}$$

Angle between vectors

Let $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ be vectors in $\mathbb{R}^2$

Inner product in 2D: $\boxed{\mathbf{a} \cdot \mathbf{b} = (a_1, a_2) \cdot (b_1, b_2) = a_1 b_1 + a_2 b_2}$

Inner product nD (so also for $n = 3$):

$$\boxed{\mathbf{a} \cdot \mathbf{b} = (a_1, \ldots, a_n) \cdot (b_1, \ldots, b_n) = a_1 b_1 + \cdots + a_n b_n}$$

Useful for:

- length: $\boxed{\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} = \sqrt{a_1^2 + \cdots + a_n^2}}$
- angle: $\boxed{\cos(\mathbf{a}, \mathbf{b}) = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|}} \Leftarrow \boxed{\underbrace{\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| \cdot |\mathbf{b}| \cdot \cos(\mathbf{a}, \mathbf{b})}_{\text{inner product rule}}}$

  Thus also $\boxed{\mathbf{a} \cdot \mathbf{b} = \cos(\alpha) \, \|\mathbf{a}\| \, \|\mathbf{b}\|, \quad \alpha = \angle(\mathbf{a}, \mathbf{b})}$

Different notation for inner product: $\boxed{\langle \mathbf{a}, \mathbf{b} \rangle}$ and $\boxed{\mathbf{a}^T \mathbf{b}}$

Angle between vectors

Ex: Calculate the angle θ between $\underbrace{\begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix}}_{\mathbf{v}}$ and $\underbrace{\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}}_{\mathbf{w}}$: Inner product rule:

$$\begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \left\| \begin{bmatrix} 1 \\ \sqrt{3} \end{bmatrix} \right\| \cdot \left\| \begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} \right\| \cdot \cos\theta$$

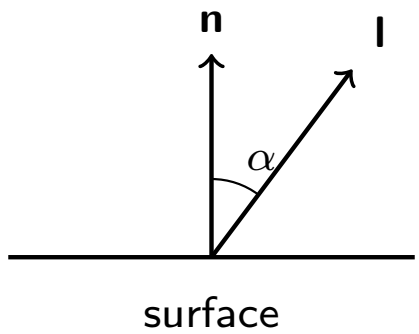
or (length of both vectors is $\sqrt{(\pm 1)^2 + (\sqrt{3})^2} = \sqrt{1+3} = 2$):

$$\underbrace{1 \cdot -1 + \sqrt{3} \cdot \sqrt{3}}_{\mathbf{v} \cdot \mathbf{w}} = \underbrace{2}_{|\mathbf{v}|} \cdot \underbrace{2}_{|\mathbf{w}|} \cdot \cos\theta$$

and $2 = 4 \cdot \cos\theta \Longrightarrow \cos\theta = \frac{1}{2} \Longrightarrow \theta = \pm\pi/3$ (solution with minus sign goes away)
(the angle between two vectors in maximal π!)

How bright is a surface under a light?

A 3D renderer needs to shade a flat surface: brighter when it faces the light, darker when it faces away.
Let $\mathbf{n}$ be the (unit-length) vector perpendicular to the surface, and $\mathbf{l}$ the (unit-length) vector pointing to the light source. Since $\|\mathbf{n}\| = \|\mathbf{l}\| = 1$:



$$\mathbf{n} \cdot \mathbf{l} = \cos(\alpha), \qquad \alpha = \angle(\mathbf{n}, \mathbf{l})$$

- Surface directly facing the light: $\alpha = 0 \Longrightarrow \mathbf{n} \cdot \mathbf{l} = 1$ (brightest)
- Surface at a grazing angle: α close to $\frac{\pi}{2} \Longrightarrow \mathbf{n} \cdot \mathbf{l}$ close to 0 (dark)
- Surface facing away: $\alpha > \frac{\pi}{2} \Longrightarrow \mathbf{n} \cdot \mathbf{l} < 0$ (no light – clamp to 0)

Brightness $\propto \mathbf{n} \cdot \mathbf{l}$ is exactly the classic Lambertian shading model used in real-time 3D graphics – a single dot product per surface, computed millions of times per frame.

Distances, angles and the inner product

- $\mathbf{x} \cdot \mathbf{x} \geq 0$

  logical, is same as $\|\mathbf{x}\|^2$, thus ≥ 0

- $\mathbf{x} \cdot \mathbf{x} = 0$ only if $\mathbf{x} = \mathbf{0}$

  only the $\mathbf{0}$-vector has length 0

- $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$

  just as for the multiplication of numbers, the order does not matter

- $(\lambda \mathbf{x}) \cdot \mathbf{y} = \lambda (\mathbf{x} \cdot \mathbf{y})$

  if $\mathbf{x}$ eg 2 times as long, then the inner product is 2 times as big

  logical, since $\mathbf{x} \cdot \mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos(\alpha)$

Orthogonal vectors

Vectors $\mathbf{a}$ and $\mathbf{b}$ are called orthogonal (or perpendicular) if $\mathbf{a} \cdot \mathbf{b} = 0$

Notation: $\mathbf{a} \perp \mathbf{b}$

Works with: $\cos(\mathbf{a}, \mathbf{b}) = \dfrac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|}$

because then the numerator is 0, so $\cos(\mathbf{a}, \mathbf{b}) = 0$, thus $\angle(\mathbf{a}, \mathbf{b}) = \frac{\pi}{2}$

Ex: $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are indeed orthogonal

but also $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

Orthogonal vectors

In $\mathbb{R}^2$ the vectors that are orthogonal to a given vector can be calculated

Ex: Which vectors are $\perp (1, 2)$?
We need to have: $(a_1, a_2) \cdot (1, 2) = a_1 + 2a_2 = 0$
Useful trick:
switch the components, and give one an extra minus sign, thus:

- $(2, -1)$, because $(2, -1) \cdot (1, 2) = 0$
- $(-2, 1)$, because $(-2, 1) \cdot (1, 2) = 0$
- and also all multiples of $(-2, 1)$ and $(2, -1)$:
  $(-2\lambda, \lambda) = \lambda(-2, 1)$ for $\lambda \in \mathbb{R}$

(In $\mathbb{R}^2$ a line of vectors is perpendicular on a given vector!)

Quiz: Vector calculus

1. Given vectors $\mathbf{v} =< -2, 3 >$ and $\mathbf{u} =< 4, 6 >$, find $\mathbf{v} + 2\mathbf{u}$ and $\mathbf{u} - 4\mathbf{u}$

2. Given vectors $\mathbf{v} =< 1, -2 >$ and $\mathbf{u} =< u_1, u_2 >$, find components of vector $\mathbf{u}$ so that $\mathbf{v} + 3\mathbf{u} = 0$

3. Given vectors $\mathbf{v} =< 4, 1 >$ and $\mathbf{u} =< u_1, u_2 >$, find components of vector $\mathbf{u}$ so that $2\mathbf{v} - 3\mathbf{u} = 0$

4. Calculate the distance between the points $A(3, 1, 7)$ and $B(5, -1, 3)$

5. Calculate the angle between the vectors $\mathbf{v} =< 1, 1, 1 >$ and $\mathbf{w} =< 4, 0, -3 >$

Quiz: Vector calculus

1. Given vectors $\mathbf{v} =< -2, 3 >$ and $\mathbf{u} =< 4, 6 >$, find
   $\mathbf{v} + 2\mathbf{u}$ and $\mathbf{u} - 4\mathbf{u}$
   $\mathbf{v} + 2\mathbf{u} =< 6, 15 >$ and $\mathbf{u} - 4\mathbf{u} =< -12, -18 >$
2. Given vectors $\mathbf{v} =< 1, -2 >$ and $\mathbf{u} =< u_1, u_2 >$, find components of vector $\mathbf{u}$ so that $\mathbf{v} + 3\mathbf{u} = 0$
   $\mathbf{u} =< -\frac{1}{3}, \frac{2}{3}$
3. Given vectors $\mathbf{v} =< 4, 1 >$ and $\mathbf{u} =< u_1, u_2 >$, find components of vector $\mathbf{u}$ so that $2\mathbf{v} - 3\mathbf{u} = 0$
   $\mathbf{u} =< \frac{8}{3}, \frac{2}{3} >$
4. Calculate the distance between the points $A(3, 1, 7)$ and $B(5, -1, 3)$
   $d = 2\sqrt{5}$
5. Calculate the angle between the vectors $\mathbf{v} =< 1, 1, 1 >$ and $\mathbf{w} =< 4, 0, -3 >$
   $\cos(\phi) = \frac{1}{5\sqrt{2}}$

Linear combination of vectors

In linear algebra, we say for vectors $\mathbf{v}_1, \ldots \mathbf{v}_n$ and real numbers $a_1, \ldots, a_n$ that the sum (-vector) $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$
is a linear combination.

Let's get to work with that.

Linear functions and combinations

A function is called linear if:

- in 1 variable: $f(x) = ax$ (or sometimes also with "$+b$")
- in 2 variables: $f(x, y) = ax + by$
- in n variables: $f(x_1, \ldots, x_n) = a_1x_1 + a_2x_2 + \cdots + a_nx_n$
- in n vectors: $f(\mathbf{v}_1, \ldots, \mathbf{v}_n) = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n$

Thus linear is always: sum of constants times variables
What is not-linear? Everything else, eq:

- everything containing multiplications of variables, eg x^2, xy, etc
- sin, cos, tan, e^x, $\ln(x)$, etc etc

Read examples [V 1.11]

Lines: An alternative to $ax + by = c$ in $\mathbb{R}^2$ and also $\mathbb{R}^3$

We have seen: Every line in $\mathbb{R}^2$ is described by:

All (x, y) for which: $ax + by = c$
Different a, b, c represent possibly different lines.
But it must not: $2x - 3y = 1$ is the same line as $3y - 2x = -1$
(multiply left/right with -1).

In $\mathbb{R}^3$

$$ax + by + cz = d$$

does not represent a line.

Lines: An alternative to $ax + by = c$ in $\mathbb{R}^2$ and also $\mathbb{R}^3$

Ex: What does $ax + by + cz = d$ describe in $\mathbb{R}^3$? Let's take an example:

$$0 \cdot x + 0 \cdot y + 1 \cdot z = 0$$

($a = b = d = 0$ and $c = 1$) – or $z = 0$. This is the xy-plane!: Table:

x	y	z	$0 \cdot x + 0 \cdot y + z$
0	0	0	$0 + 0 + 0 = 0$
1	0	0	$0 \cdot 1 + 0 + 0 = 0$
0	1	0	$0 + 0 \cdot 1 + 0 = 0$
1	1	0	$0 \cdot 1 + 0 \cdot 1 + 0 = 0$
$\vdots$	$\vdots$	0	$0 = 0$

The 4 points $(0, 0, 0)$, $(0, 1, 0)$, $(1, 0, 0)$ and $(1, 1, 0)$ are the corners of a square in the xy-plane. It cannot be a straight line!
We need: **An expression for a line that works in $\mathbb{R}^2$, $\mathbb{R}^3$, $\mathbb{R}^4$ ...**

Parameter representation of a Line

Let $\mathbf{a} = \begin{bmatrix} x_A \\ y_A \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} x_B \\ y_B \end{bmatrix}$ be two coordinate vectors of the points A and B. (the same for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$). It holds that

$$\mathbf{x} = \mathbf{x}(\lambda) = \mathbf{a} + \lambda\,(\mathbf{b} - \mathbf{a}) \text{ for all } \lambda \in \mathbb{R}$$

describes all points on the line through A and B. Specifically

- $\lambda = 0$: $\mathbf{x} = \mathbf{a} + 0 \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{a} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{a}$
- $\lambda = 1$: $\mathbf{x} = \mathbf{a} + 1 \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{a} + \mathbf{b} - \mathbf{a} = \mathbf{b}$

The representation $\mathbf{x} = \mathbf{x}(\lambda) = \mathbf{a} + \lambda\,(\mathbf{b} - \mathbf{a})$ with the parameter λ is called parameter representation or vector representation. Definitions:

- $\mathbf{a}$ is the support vector or the start point
- $\mathbf{b} - \mathbf{a}$ is the direction vector

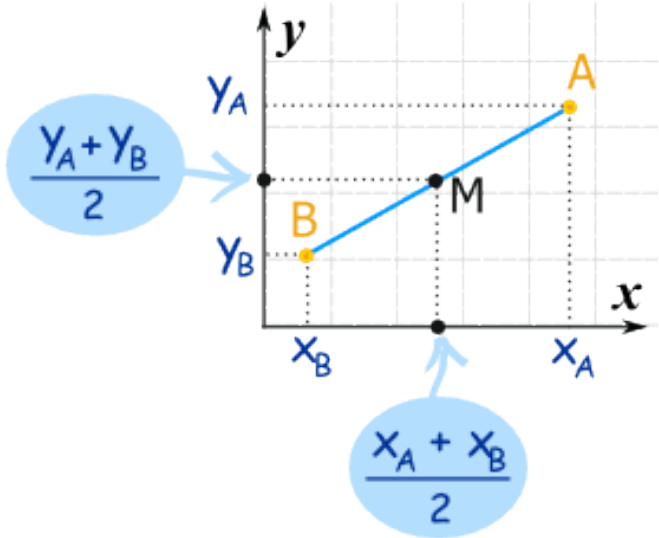
Parameter representation of a Line, now in $\mathbb{R}^3$

Ex: Parameter representation of a line through points $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$:

$$\begin{aligned}
\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 2-1 \\ 3-1 \\ 1-1 \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \lambda \underbrace{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}}_{\text{direction vector}} \\
&= \begin{bmatrix} 1+\lambda \\ 1+2\lambda \\ 1+0\cdot\lambda \end{bmatrix} = \begin{bmatrix} 1+\lambda \\ 1+2\lambda \\ 1 \end{bmatrix}
\end{aligned}$$

Lines, line segments, straight lines

For all $\lambda \in \mathbb{R}$ is $\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$ on the line through A en B:



- $\lambda = 0$: $\mathbf{x} = \mathbf{a} + 0 \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{a} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \mathbf{a}$ (point A)
- $\lambda = 1$: $\mathbf{x} = \mathbf{a} + 1 \cdot (\mathbf{b} - \mathbf{a}) = \mathbf{a} + \mathbf{b} - \mathbf{a} = \mathbf{b}$ (point B)
- $\lambda = \frac{1}{2}$: $\mathbf{x} = \mathbf{a} + \frac{1}{2} \cdot (\mathbf{b} - \mathbf{a}) = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ middle between $\mathbf{a}$ en $\mathbf{b}$
- $\lambda = \frac{1}{4}$: $\mathbf{x} = \frac{3}{4}\mathbf{a} + \frac{1}{4}\mathbf{b}$ middle between M and A.

Complete lines, half lines and line segments

Parameter representation $\boxed{\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})}$

yields for each $\lambda \in \mathbb{R}$
one point $\mathbf{x}$ on the line through points B and A
Depending on what λ can be, this is a line segment, line etc:

- $\lambda \in [0, 1]$: line segment between A and B including these points
- $\lambda \geq 0$: half line from A through B and further
- $\lambda \in \mathbb{R}$: (complete) line through points A and B

Note that $\mathbf{a} + \lambda(\mathbf{b} - \mathbf{a})$ is a linear combination because
$\mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) = \mathbf{a} + \lambda\mathbf{b} - \lambda\mathbf{a} = \lambda\mathbf{b} + (1 - \lambda)\mathbf{a}$ is of the form: scalar times vector plus scalar times vector.

Lines, line segments, straight lines

Ex: A line has several parameter representations:

Parameter representations are not unique! $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} -2 \\ -4 \\ 0 \end{bmatrix}$ just take $\mu = -\frac{1}{2}\lambda$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} -2 \\ -4 \\ 0 \end{bmatrix}$ just take $\mu = -\frac{1}{2}\lambda - \frac{1}{2}$

If your neighbor gets a different answer, it does not mean anything! ...

But: All direction vectors are "the same" except for a minus sign.

Lines in $\mathbb{R}^2$: Function vs. parameter representation

Change from function to parameter representation:

Ex: $y = 2x + 3$ goes through $(0, 3)$ and $(1, 5)$, thus

parameter representation $\begin{bmatrix} 0 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \lambda \\ 3 + 2\lambda \end{bmatrix}$

Ex: Let a parameter representation be $\begin{bmatrix} 1 + 2\lambda \\ 2 + 3\lambda \end{bmatrix}$

then the line goes through $(1, 2)$ (for $\lambda = 0$) and $(3, 5)$ (for $\lambda = 1$)

thus it's the equation $y = \frac{3}{2}x + \frac{1}{2}$

Lines in $\mathbb{R}^3$

Can be given as in $\mathbb{R}^2$ by parametrisation:
starting point and direction vector

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \lambda \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

A line can also be given by the intersection of two planes

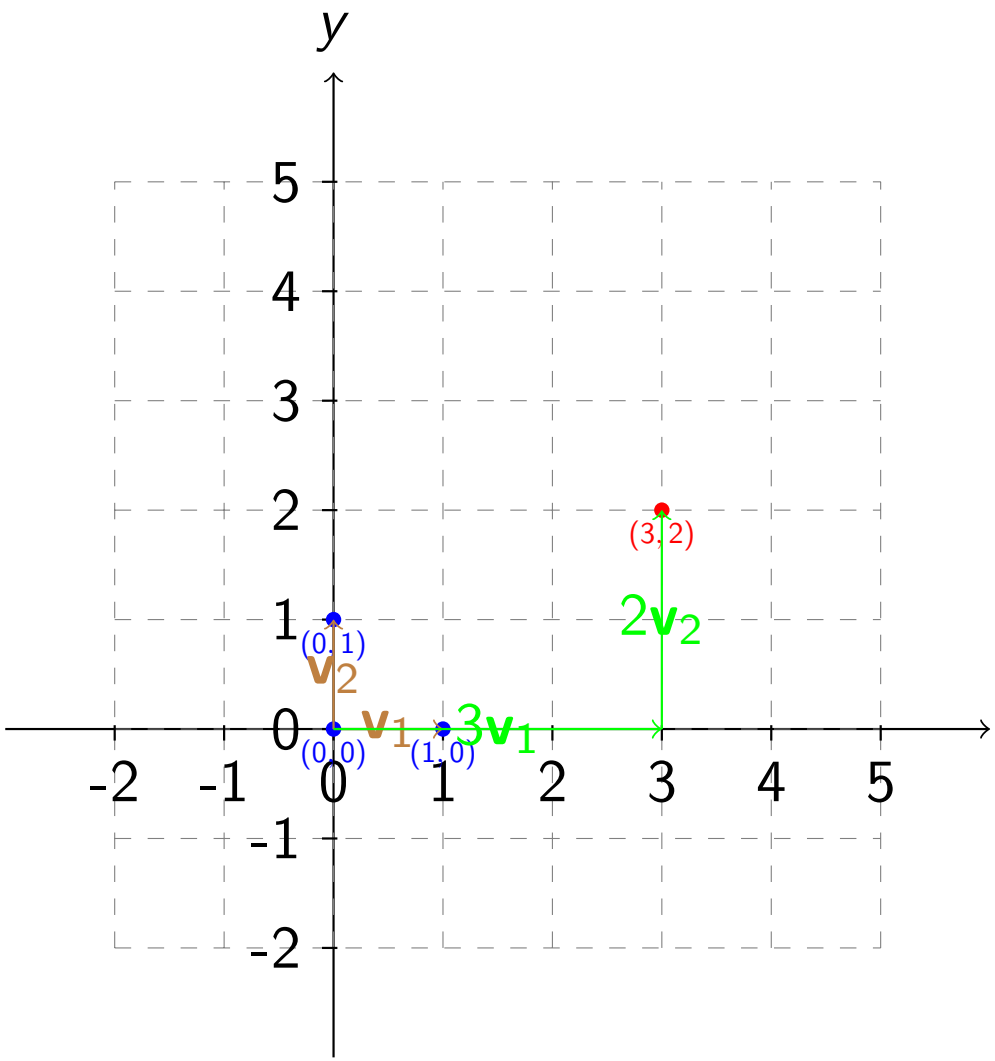
Lines

- Determine a parametric representation of the lines through:
  - $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 5 \\ -1 \\ 4 \end{bmatrix}$
  - $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$
- Give an equation for the line:

  $\mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ -1 \end{bmatrix}$
- Find a parametric representation for the following lines in $\mathbb{R}^2$

  $3x - 4y + 7 = 0$

Vectors in $\mathbb{R}^2$



start point $\mathbf{p} = (0, 0)$, to reach $(3, 2)$

direction $\mathbf{v}_1 = (1, 0) - (0, 0) = (1, 0)$

direction $\mathbf{v}_2 = (0, 1) - (0, 0) = (0, 1)$

way to $\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \dots$

way to $\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

way to $\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \mathbf{p} + 3\mathbf{v}_1 + 2\mathbf{v}_2$

xy-plane: Each point (x, y) determined by unique λ, μ: $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

xy-plane: $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{p} + \lambda\mathbf{v}_1 + \mu\mathbf{v}_2$

Why the parameter representation?

The 2D notation $y = ax + b$ does not work for equations in the 3D space!

Ex: What object is $x - y + z = 1$ in $\mathbb{R}^3$?
Answer: That depends on the set of points (x, y, z) that satisfies the equation.
Let's try to find a description for that:

- 1 equation with 3 unknowns. Chose 2 arbitrarily:
- $z = \lambda$ and $y = \mu$ and calculate x:
- $x = 1 - \lambda + \mu$

So for all $\lambda, \mu \in \mathbb{R}$ the resulting **x** satisfies $x - y + z = 1$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 - \lambda + \mu \\ \mu \\ \lambda \end{bmatrix}$$

Why the parameter representation?

Different notation:

$$\begin{aligned}
\begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 1 - \lambda + \mu \\ \mu \\ \lambda \end{bmatrix} \\
&= \begin{bmatrix} 1 + (-1) \cdot \lambda + 1 \cdot \mu \\ 0 + 0 \cdot \lambda + 1 \cdot \mu \\ 0 + 1 \cdot \lambda + 0 \cdot \mu \end{bmatrix} \\
&= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}
\end{aligned}$$

represents all linear combinations of 2 directions from 1 support point.
Such an object is always a plane!
Summarized:

- in 2D: $ax + by = c$ is the equation of a line
- in 3D: $ax + by = c \iff ax + by + 0 \cdot z = c$ is the equation of a plane!

Planes in $\mathbb{R}^3$

Planes can be given by a parametrisation:

start point and 2 direction vectors:

$$\mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) + \mu(\mathbf{c} - \mathbf{a}) \text{ for all } \lambda, \mu \in \mathbb{R}$$

λ and μ are the parameters

$$\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} + \lambda \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \\ z_2 - z_1 \end{bmatrix} + \mu \begin{bmatrix} x_3 - x_1 \\ y_3 - y_1 \\ z_3 - z_1 \end{bmatrix}$$

Ex: the xy-plane: $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

The parameter representation of the xy-plane

Addition yields $\begin{bmatrix}0\\0\\0\end{bmatrix} + \lambda \begin{bmatrix}1\\0\\0\end{bmatrix} + \mu \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}\lambda\\\mu\\0\end{bmatrix}$ that the points $(\lambda, \mu, 0)$ are described for $\lambda, \mu \in \mathbb{R}$.

Ex: The equation $z = 0$ describes the same xy-plane:

- 1 equation with 3 unknowns: $0 \cdot x + 0 \cdot y + z = 0$. Choose 2 arbitrarily:
- $x = \lambda$ and $y = \mu$ and calculate z:
- $z = 0$.

So that for all $\lambda, \mu \in \mathbb{R}$ the resulting **x** satisfy $z = 0$:

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}\lambda\\\mu\\0\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix} + \lambda \begin{bmatrix}1\\0\\0\end{bmatrix} + \mu \begin{bmatrix}0\\1\\0\end{bmatrix}$$

The parameter representation is not unique

Take V the xy-plane:

$$V: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \lambda \\ \mu \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \lambda \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\mathbf{v}} + \mu \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{w}}$$

All points $P = (a, b, 0)$ are in V:

$$V: \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We obtain the point P by choosing $\lambda = a$ and $\mu = b$

The parameter representation is not unique

Take V the xy-plane:

$$V: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \lambda \\ \mu \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \lambda \underbrace{\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}}_{\mathbf{r}} + \mu \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\mathbf{s}}$$

All points $P = (a, b, 0)$ are in V:

$$V: \begin{bmatrix} a \\ b \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \frac{1}{2}a \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + (b + \frac{1}{2}a) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

We obtain the point P by choosing $\lambda = \frac{1}{2}a$ and $\mu = b + \frac{1}{2}a$
A plane can be described by many different directions
Here $\mathbf{r} = 2\mathbf{v} - \mathbf{w}$ and $\mathbf{s} = \mathbf{w}$

Orthogonal vectors

In $\mathbb{R}^3$ a plane of vectors is orthogonal to a given vector.

Ex: Which vectors $\mathbf{v} \in \mathbb{R}^3$ are perpendicular on $\begin{bmatrix} -3 \\ 4 \\ 1 \end{bmatrix}$? For such a vector

$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$ it must hold: $\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \cdot \begin{bmatrix} -3 \\ 4 \\ 1 \end{bmatrix} = 0 \iff -3v_x + 4v_y + v_z = 0$

- 1 equation with 3 unknown. Choose 2 arbitrarily:
- $v_x = \lambda$ and $v_y = \mu$ and calculate v_z:
- $v_z = 3\lambda - 4\mu$

which is a plane through the origin:

$$\begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} \lambda \\ \mu \\ 3\lambda - 4\mu \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \\ -4 \end{bmatrix}$$

Orthogonal vectors

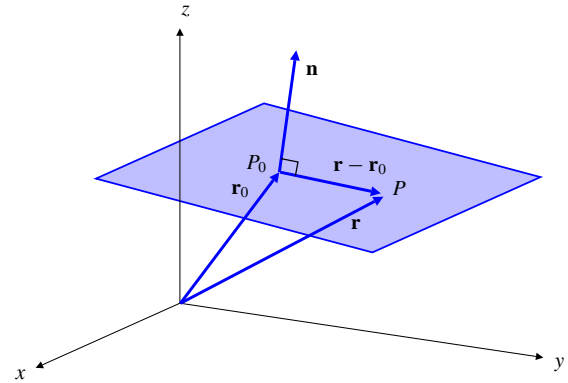
Ex: For which numbers $a \in \mathbb{R}$ is $\begin{bmatrix} a \\ 4 \\ 1 \end{bmatrix}$ orthogonal to $\begin{bmatrix} a \\ a \\ 3 \end{bmatrix}$?

We must have:

$$\begin{aligned} (a, 4, 1) \cdot (a, a, 3) &= 0 \\ \iff a^2 + 4a + 3 &= 0 \\ \iff (a+1)(a+3) &= 0 \\ \iff a = -1 \vee a &= -3 \end{aligned}$$

Planes and normals (see also [A 10.4])

Let $P_0 = (x_0, y_0, z_0)$ be an arbitrary point in the plane
and $\mathbf{n} = (n_1, n_2, n_3)$ a normal(vector) on this plane (length does not matter):



normal vector = vector that is orthogonal to the plane

The plane consists of all points $P = (x, y, z)$ for which it holds:
$(x - x_0,\ y - y_0,\ z - z_0)$ is orthogonal to $\mathbf{n}$
Thus for (x, y, z) in the plane:

$$0 = \mathbf{n} \cdot (x - x_0, y - y_0, z - z_0) = n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0)$$

So $n_1 x + n_2 y + n_3 z = \underbrace{n_1 x_0 + n_2 y_0 + n_3 z_0}_{d}$

Normal vector form of a plane: $\mathbf{n} \cdot \mathbf{x} = d$

Planes and normals

Earlier, we saw that the plane V with equation $x - y + z = 1$ has the vector representation:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \mu \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \qquad \text{and normal } \mathbf{n} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

The normal $\mathbf{n}$ must be orthogonal to both directions V!

Check:

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 1 \cdot -1 + -1 \cdot 0 + 1 \cdot 1 = 0 \text{ works}$$

$$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 \cdot 1 + -1 \cdot 1 + 1 \cdot 0 = 0 \text{ works too}$$

Planes and normals

The plane V through point $P = (1, 2, 3)$ with normal $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$?

has the equation $n_1 x + n_2 y + n_3 z = d$

Thus $-x + 0 \cdot y + z = d \Longrightarrow -x + z = d$

Point P lies in V: Filling in yields $-1 + 3 = 2 = d$.

Planes and normals

The plane V through point $P = (1, 2, 3)$ with normal $\mathbf{n} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$?

has the equation $-x + z = 2$ (previous slide)

But also the equation $-2x + 2z = 4$ with normal $\mathbf{n} = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}$

and equation $x - z = -2$ with normal $\mathbf{n} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

The normals: All parallel!

Equations for a plane and normal on plane

$$n_1x + n_2y + n_3z = \underbrace{n_1x_0 + n_2y_0 + n_3z_0}_{d}$$

Formula for d unimportant, but:

- From the equation of the plane $n_1x + n_2y + n_3z = d$ you can see:
  normal $= (n_1, n_2, n_3)$
- And vice versa: from the normal $= (n_1, n_2, n_3)$ you know the equation of the plane: $n_1x + n_2y + n_3z = d$ where you still need to determine d via filling in 1 point P in the plane

Lines in $\mathbb{R}^3$

Reminder: 2 options: given by

- parameter representation
- or 2 equations: intersection of 2 planes

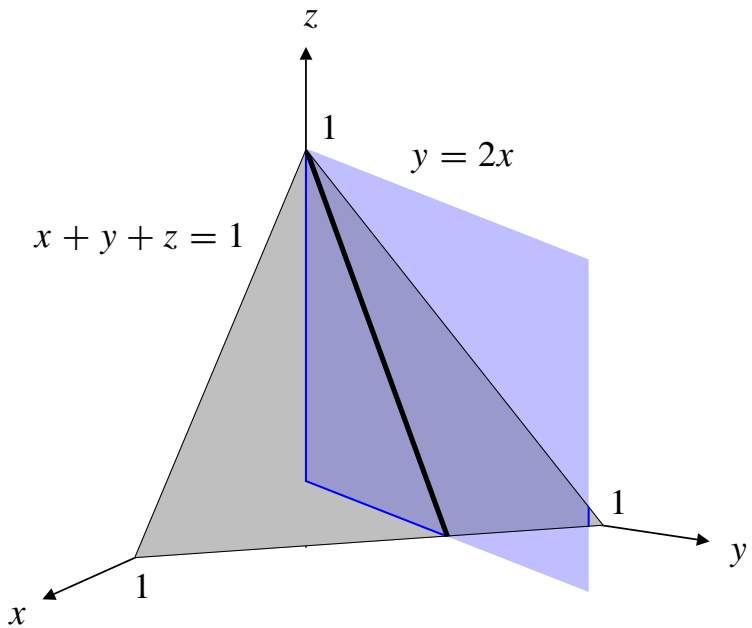
Parameter representation: $(x, y, z) = (x_1, y_1, z_1) + t\,(a, b, c), \quad t \in \mathbb{R}$
support point + $t \cdot$ direction vector

Ex: $\begin{aligned} x &= 2+t \\ y &= 3 \\ z &= -4t \end{aligned}$ is rewritten as $(x, y, z) = (2, 3, 0) + t\,(1, 0, -4)$

Ex: Line through $(1, -2, 3)$ orthogonal to plane $x - 2y + 4z = 5$ has parameter representation
$(x, y, z) = (1, -2, 3) + t\,(1, -2, 4) = (1 + t,\ -2 - 2t,\ 3 + 4t)$

Lines in $\mathbb{R}^3$

Also possible by 2 equations: intersection of 2 planes



One can make a parameter representation out of this by finding 2 points, eg:
choose $x = 0$ then $y = 0$ and $z = 1$
choose $x = 1$ then $y = 2$ and $z = -2$
A parameter representation is then
$(0, 0, 1) + t\,((1, 2, -2) - (0, 0, 1)) = (0, 0, 1) + t\,(1, 2, -3)$

Planes

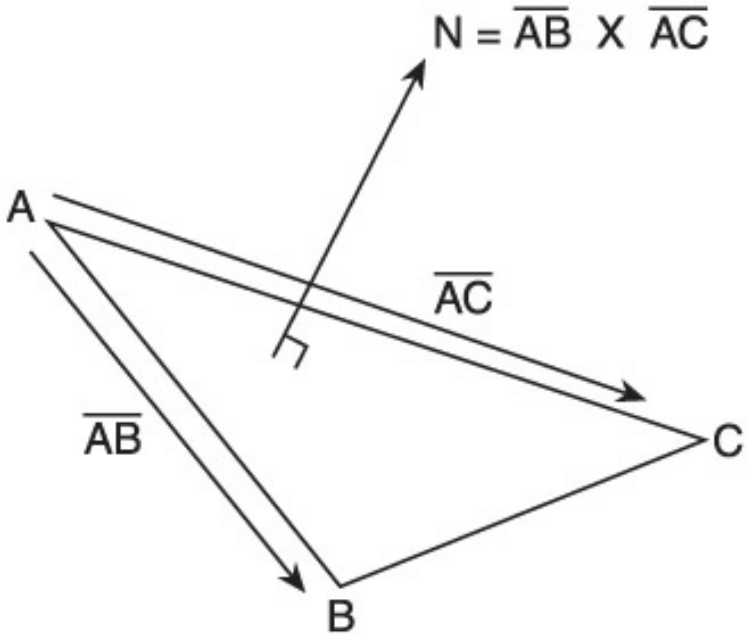
- Give a parameter representation of the line through $(5, -2, -5)$ parallel to a vector $(7, 7, 10)$
- The same for the line through $(-8, -7, 2)$ orthogonal to the plane $3x + 3y + 3z = 8$
- Give the equation for the plane through $(3, 5, 5)$ and normal to the vector $(8, 6, 9)$
- Where does the line $x = -8 + 3t, \;\; y = 4 + 7t, \;\; z = -9 - 8t$ intersect the plane $9x - 6y + 3z = -11$

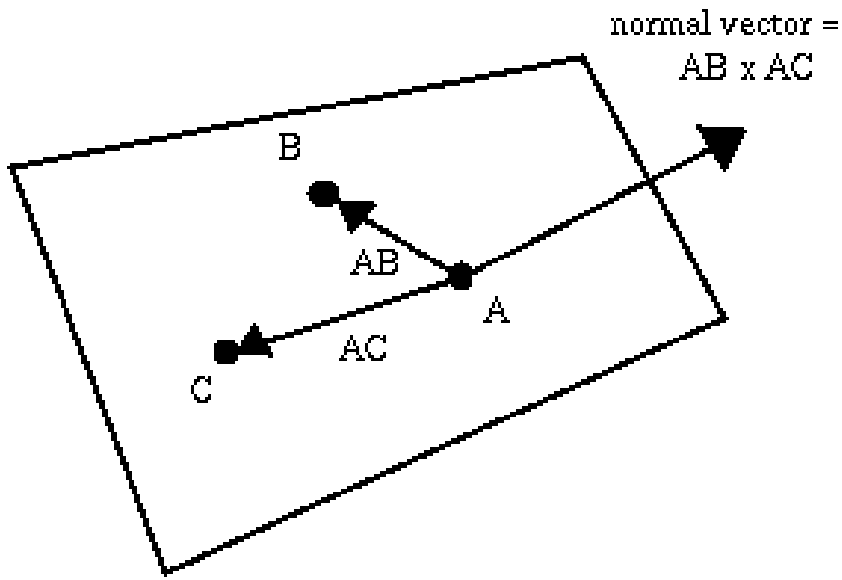
Line in $\mathbb{R}^2$ and plane in $\mathbb{R}^3$

Note the similarities and differences:
Line in $\mathbb{R}^2$: $ax + by = c$
(a, b, c fixed, x, y variable)
Plane in $\mathbb{R}^3$: $ax + by + cz = d$
(a, b, c, d fixed, x, y, z variable)

The cross product vector

Cross product vector: A vector orthogonal to two different vectors.
Three points A, B and C define a start point (eg A) and two directions from A ($\vec{AB} = \mathbf{b} - \mathbf{a}$ and $\vec{AC} = \mathbf{c} - \mathbf{a}$) and with that a plane with vector representation: $\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) + \mu(\mathbf{c} - \mathbf{a})$.
The normal N (vector $\mathbf{n}$) is orthogonal to both vectors.





and can be calculated with the so called cross product.

Definition of the cross product

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix}$$

Ex: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Ex: $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

Ex: $\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$

Ex: $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$

What is noteworthy about these examples for $\mathbf{v} \times \mathbf{w}$?

- Cross product is orthogonal to the 2 vectors $\mathbf{v}$ and $\mathbf{w}$
- Length is the same as the surface of a parallellogram formed by $\mathbf{v}$ and $\mathbf{w}$

This always turns out to be true.

Cross product and normal vector

The cross product is useful for calculating the normal vector on the plane

Ex: Plane with parametrisation $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \mu \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

this is the (x, y)-plane

Normal is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Cross product: properties

- $\mathbf{x} \times \mathbf{x} = \mathbf{0}$

  cross product of a vector with itself is the $\mathbf{0}$-vector

  logical, because then the parallellogram has the surface 0

- $(\mathbf{x} \times \mathbf{y}) \cdot \mathbf{x} = 0$

  cross product of one vector with another is perpendicular to the other

  this is 1 of the 2 basic properties of the cross product

- $\mathbf{y} \times \mathbf{x} = -(\mathbf{x} \times \mathbf{y})$

  when you swap the order, you get the opposite direction

  "corkscrew-rule"

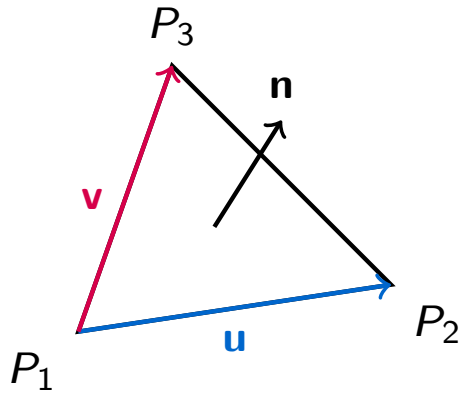
- $(\lambda \mathbf{x}) \times \mathbf{y} = \lambda (\mathbf{x} \times \mathbf{y})$

  if $\mathbf{x}$ is eg 2 times longer, then the cross product is 2 times as long

  logical, because the length of the cross product is surface of a parallellogram

Which way does a 3D surface face?

3D graphics represents every surface as a mesh of tiny triangles. To shade a triangle correctly (recall the dot-product lighting formula from last week), the renderer first needs the vector perpendicular to the triangle – its normal.



Given a triangle with corners P_1, P_2, P_3, take two of its edge vectors

$$\mathbf{u} = P_2 - P_1, \qquad \mathbf{v} = P_3 - P_1$$

The normal is simply

$$\boxed{\mathbf{n} = \mathbf{u} \times \mathbf{v}}$$

exactly the same computation as the plane-normal example on the previous slide.

Every triangle in every 3D model you have ever seen carries a cross-product-computed normal, used both for lighting and for deciding which side of the surface is the "front" (visible) side.

Distance of a point to a plane

Distance of point (p, q, r) to a plane $ax + by + cz = d$

1. Set up the eq. for the line through point orthogonal to plane:
   $(x, y, z) = (p, q, r) + \lambda (a, b, c)$
2. Calculate the intersection point of the line with the plane: fill in $x = p + \lambda a, \quad y = q + \lambda b, \quad z = r + \lambda c$ into the plane equation
3. Calculate distance between (p, q, r) and intersection point

Distance of a point to a line

Distance of the point to a line $\mathbb{R}^3$

1. Set up the eq. of a plane through point orthogonal to the line;
   $ax + by + cz = d$, with (a, b, c) the direction vector of the line
2. Calculate d by filling in the point
3. calculate intersection of line with plane
4. calculate the distance between point and intersection

Surface of a triangle in plane and space

Surface triangle (a, b, c), (d, e, f), (g, h, i) in $\mathbb{R}^3$

1. calculate cross product of $(d, e, f) - (a, b, c)$ and $(g, h, i) - (a, b, c)$
2. surface $= \frac{1}{2} \cdot$ length of this vector

Surface triangle (a, b), (c, d), (e, f) in $\mathbb{R}^2$

1. Calculate cross product of $(c, d, 0) - (a, b, 0)$ and $(e, f, 0) - (a, b, 0)$
2. surface $= \frac{1}{2} \cdot$ length of this vector

Note: there are also other methods you can use.

When inner product/when cross product

Let $\mathbf{u}, \mathbf{v}$ be two vectors.

Inner product:

- To see if $\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$
- Calculation of the angle θ between $\mathbf{u}$ and $\mathbf{v}$: $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta$

Cross product:

- Calculation of a vector $\mathbf{n}$ orthogonal to $\mathbf{u}$ and $\mathbf{v}$: $\mathbf{n} = \mathbf{u} \times \mathbf{v}$
- To see if $\mathbf{u} \parallel \mathbf{v} \iff \mathbf{u} \times \mathbf{v} = 0$
- Calculation of the surface O of the parallellogram spanned by $\mathbf{u}$ and $\mathbf{v}$: $O = |\mathbf{u} \times \mathbf{v}|$

Week 2: We have seen

- Trigonometric functions sin, ...
- Powers e^x 2^x, ...
- Vectors in 2 and 3 dimensions
- Lines and planes
- Distance, angle, inner product
- Cross product

